

MB 573 : Cloud Computing for Business

Course Code : MB 573

Course Name : Cloud Computing for Business

L-T-P-C : 2-0-0-2

Intended for : MBA

Prerequisite :

Mutual Exclusion : NA

Approval: 54th BoA

Course Contents

- **Intorduction to Cloud Computing:** Understanding what Computing is, Trends in Computing, Centralized vs Distributed Computing, Soft introduction to Grid, Cluster and Utility Computing, Why Cloud Computing, Introduction to Cloud Computing, Definition of Cloud, Component and Implementation of Cloud, Evolution of Cloud Computing, Cloud Characteristics, Advantages and Disadvantages of Cloud computing, Essentials, Benefits, Business and IT perspective (5 Hours)
- **Cloud Architecture and Models:** Cloud Architecture, Layered, NIST Cloud Computing Reference Architecture, Cloud Models- Service and Deployment, Cloud Service Models- IaaS, PaaS and SaaS, Cloud Service Models- Public Clouds, Private Cloud, Hybrid Cloud, Community Cloud. Architectural Design Challenges, Business applications (appropriate case studies, use cases and situation analysis). (6 Hours)
- **Cloud Storage:** Storage as-a-Service, Advantages of Cloud Storage, Cloud Storage Providers, Business applications and use cases - AWS, Google App Engine, Microsoft Azure (6 Hours)
- **Security in Cloud Environment:** Cloud Security Causes- Loss, Lack and Multitenancy, Taxonomy and Cloud Threat Models, Cloud Infrastructure Security, Security Boundaries in Cloud, Cloud Security Management Frameworks, Security -as-a-Service, Cloud Security Controls (4 Hours)
- **Cloud Virtualization and Adoption:** Cloud and Virtualization, Basics of Virtualization, Types of Virtualizations, Virtualization Defined, Virtualization Benefits, Cloud Pricing Models; Pay-as-you-go, Reserved Instances, Spot pricing, Use Cases, Cloud computing transition and adoption in Business applications (appropriate case studies and use cases). (7 Hours)

Textbooks:

1. Erl, Thomas, Ricardo Puttini, and Zaigham Mahmood, **Cloud Computing: Concepts, Technology & Architecture**, Pearson, 2013
2. Bahga, Arshdeep, and Vijay Madisetti, **Cloud Computing: A Hands-on Approach**, CreateSpace Independent Publishing Platform, 2013

References:

1. Rittinghouse, John W., and James F. Ransome, **Cloud Computing: Implementation, Management and Security**, CRC Press.



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2. Rajkumar Buyya, Christian Vecchiola, S. ThamaraiSelvi, **Mastering Cloud Computing**, Tata Mcgraw Hill.
 3. Toby Velte, Anthony Velte, Robert Elsenpeter, **Cloud Computing – A Practical Approach**, Tata Mcgraw Hill.